

S.No. : 174

NCS4102

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 4246

Roll
No.

1	1	1	1	1	1	1	1	1	1
---	---	---	---	---	---	---	---	---	---

B. Tech Examination 2025-26

Odd Semester

BASICS OF ARTIFICIAL INTELLIGENCE

Time : Three Hours]

[Maximum Marks :60

Note :- Attempt all questions.

SECTION –A

1. Attempt all parts of the following : $1 \times 8 = 8$

- (a) Describe the turing test for intelligence.
- (b) What is heuristic function?
- (c) What are the types of nodes in decision tree?
- (d) Justify the use of searching in game.
- (e) Explain the global search algorithm.

[P. T. O.]

- (f) Define reinforcement learning.
- (g) Define the term knowledge. What is the role of knowledge in artificial intelligence?
- (h) Write the difference between the propositional and predicate logic.

SECTION – B

2. Attempt any two parts of the following : $6 \times 2 = 12$

- (a) Define artificial intelligence. Explain the techniques and characteristics of artificial intelligence.
- (b) Explain depth first search algorithm with a suitable example.
- (c) Define agent program. Explain the following agent programs with respect to intelligent systems.
 - (i) Simple Reflex Agents
 - (ii) Model-Based Reflex Agents

- (d) List and explain the applications of artificial intelligence.

SECTION – C

Note:- Attempt all questions. Attempt any two parts from each question. $5 \times 8 = 40$

- 3. (a) Define PEAS representation. Give the PEAS representation of a self-driving car.
 - (b) Discuss in detail about the A* algorithm using a suitable example.
 - (c) Explain backward chaining with an example.
- 4. (a) Explain fuzzy system and its applications.
 - (b) How a problem is formally defined? List down the components of it.
 - (c) What are predicates? Explain its syntax and semantics.
- 5. (a) Describe the N-queen problem in detail. Also explain with an example.

[P. T. O.]

- (b) What is knowledge-based agent? Explain its role and importance.
 - (c) What is a Bayesian network, and why is it important in AI?
6. (a) What are the types of artificial intelligence? Explain with example.
- (b) What are the ethical concerns about AI system?
 - (c) Describe the rules of inference in first-order-predicate logic with suitable examples.
